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Manufacturing manuals: index

Start with **STANDARD_PROCEDURES.md** (PCB ordering, SMT/THT assembly, soldering, programming, 3D printing, laser cutting, QA, packaging, compliance, troubleshooting) and the templates. Each product has a `MANUFACTURING_MANUAL.md` with its bill of materials, build steps, calibration, end-of-line test and limits.

Honest status: nothing in this repository has been built, assembled or tested as hardware. 'Order-ready' means the automatic checks pass (KiCad DRC 0 errors, schematic equals layout, every part has a verified LCSC number, Gerber/BOM/CPL consistent): it does **not** mean an engineer has reviewed the layout. Treat the first batch as a prototype run.

Utility tools

Project	Manual	Board	Main caveat
Seeti Counter	utility10/01-seeti-counter/MANUFACTURING_MANUAL.md	**order-ready**	firmware compiled, not run on hardware
Gas Taraju (LPG cylinder scale)	utility10/02-gas-taraju/MANUFACTURING_MANUAL.md	**order-ready**	load-cell calibration per unit
Fridge Door Guard	utility10/03-fridge-door-guard/MANUFACTURING_MANUAL.md	**order-ready**	
Bijli Meter Reader	utility10/04-bijli-meter-reader/MANUFACTURING_MANUAL.md	**order-ready**	
Bijli Log	utility10/05-bijli-log/MANUFACTURING_MANUAL.md	**order-ready**	OLED/RTC drivers untested
Chulha Timers	utility10/06-chulha-timers/MANUFACTURING_MANUAL.md	**order-ready**	display driver untested
Rain Alert Bell + plate	utility10/07-rain-alert/MANUFACTURING_MANUAL.md	**order-ready**	plate: `rainplate` board
Tap Litre Meter	utility10/08-tap-litre-meter/MANUFACTURING_MANUAL.md	**order-ready**	needs bucket calibration
Masala Rack (laser-cut)	utility10/09-masala-rack/MANUFACTURING_MANUAL.md	no PCB	kerf coupon first
Laptop Stand (laser-cut)	utility10/10-laptop-stand/MANUFACTURING_MANUAL.md	no PCB	stability limit stated

Products

Project	Manual	Board	Main caveat
1 Desi Games Studio (print)	products/01-desi-games/MANUFACTURING_MANUAL.md	no PCB	language + fact review before print
2 Kalakriti Automata (laser)	products/02-automata/MANUFACTURING_MANUAL.md	no PCB	frame layouts provisional
3 Prayog STEM lab kit	products/03-prayog/MANUFACTURING_MANUAL.md	**order-ready**	sensors uncalibrated

Project	Manual	Board	Main caveat
4 Jal-Rakshak tank controller	products/04-jal-rakshak/MANUFACTURING_MANUAL.md	no PCB	**mains version cannot be made**: low-voltage prototype only
5 Kheti-Mitra irrigation	products/05-kheti-mitra/MANUFACTURING_MANUAL.md	**order-ready**	field trial required
6 Gas-Suraksha leak alarm	products/06-gas-suraksha/MANUFACTURING_MANUAL.md	no PCB	**not certified, not for sale**

Toys (10)

Project	Manual	Board	Main caveat
01 Reflex Raja	toys10/01-reflex-raja/MANUFACTURING_MANUAL.md	**order-ready**	shares the Desi Arcade board
02 Taal Memory	toys10/02-taal-memory/MANUFACTURING_MANUAL.md	**order-ready**	shares the Desi Arcade board
03 Aloo Bomb	toys10/03-aloo-bomb/MANUFACTURING_MANUAL.md	**order-ready**	shares the Desi Arcade board
04 Beat Runner	toys10/04-beat-runner/MANUFACTURING_MANUAL.md	**order-ready**	
05 Rangoli Spiro Gears	toys10/05-rangoli-spiro/MANUFACTURING_MANUAL.md	no PCB	kerf test first
06 Paper Rocket Launcher	toys10/06-paper-rocket/MANUFACTURING_MANUAL.md	no PCB	projectile toy: lab review before sale; no drawings
07 Tilt Maze	toys10/07-tilt-maze/MANUFACTURING_MANUAL.md	no PCB	gimbal frame drawing missing
08 Rubber-band Racer	toys10/08-band-racer/MANUFACTURING_MANUAL.md	no PCB	provisional cut file
09 Marble Cricket	toys10/09-marble-cricket/MANUFACTURING_MANUAL.md	no PCB	provisional cut file; calibration drop
10 Katha Cubes	toys10/10-katha-cubes/MANUFACTURING_MANUAL.md	no PCB	illustrations and proofreading missing

Bio / agri / civil / robotics programmes

Project	Manual	Board	Main caveat
KINETIC motion controller + vibration puck	kinetic/MANUFACTURING_MANUAL.md	**order-ready**	never run on a motor; AI on synthetic data
MITTI-NETRA soil + structure node	mitti-netra/MANUFACTURING_MANUAL.md	**order-ready**	modem is a plug-in breakout
BioTray vertical-farm controller	biotray/MANUFACTURING_MANUAL.md	**order-ready**	calibration values are placeholders
BioWatt MFC harvester	biowatt/MANUFACTURING_MANUAL.md	**order-ready**	cold start needs 600 mV; marginal on one MFC
GEONODE foundation monitoring	geotech/MANUFACTURING_MANUAL.md	no PCB	synthetic data; reuses MITTI-NETRA board
AQUAPONICS controller	aquaponics/MANUFACTURING_MANUAL.md	no PCB	runs on BioTray; biological limits are placeholders
SORTER bio-robotic platform	sorter/MANUFACTURING_MANUAL.md	no PCB	vision baseline only; uses KINETIC

Project	Manual	Board	Main caveat
AGROVER solar rover	agrover/MANUFACTURING_MANUAL.md	no PCB	dead-reckoning only; safety review needed

Other projects

Project	Manual	Board	Main caveat
Jal-Netra (engineering marvel)	jal-netra/MANUFACTURING_MANUAL.md	**order-ready**	**Gate 1 acoustic test first; not sellable**
ASTRADAND gesture wand (sci-fi project)	astradand/MANUFACTURING_MANUAL.md	**order-ready**	software-verified; never built; Hindi names unreviewed
Gravity toys (5)	gravity-toys/MANUFACTURING_MANUAL.md	no PCB	drawings partial
Wave 1 electronics toys	electronics-toys/wave1/MANUFACTURING_MANUAL.md	no PCB	no PCB: not manufacturable yet
Prototype packs (10)	prototypes/MANUFACTURING_MANUAL.md	no PCB	bench prototypes
DJ / party system	dj-setup/MANUFACTURING_MANUAL.md	no PCB	plan only: not manufacturable